USN

[22AIMLDE21]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER - 2023

APPLICATIONS OF AI

Time: 3 Hours Maximum **Marks: 100**

Instructions: Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**05 X 02=10**		
1.	**a.**	What is Artificial Intelligence?	L-1	CO1	**02**
	b.	What is meant by Precision medicine?	L-2	CO3	**02**
	c.	Mention any three AI applications in the field of agriculture.	L-3	CO4	**02**
	d.	What are the merits and demerits of using AI applications?	L-3	CO4	**02**
	e.	Mention the AI components used in food processing.	L-2	CO5	**02**
		PART-B	**06 X 05=30**		
2.	**a.**	Write a short note on different types of AI with its applications.	L-2	CO1	**05**
		OR			
	b.	In what ways does AI can help in daily activities? Explain in detail.	L-2	CO2	**05**
3.	**a.**	Explain the procedure for using AI to detect diseases in plants.	L-3	CO3	**05**
		OR			
	b.	Explain the procedure to classify rice grains using neural networks	L-3	CO3	**05**
4.	**a.**	Explain the concept of AI-enabled health data management techniques.	L-3	CO4	**05**
		OR			
	b.	Write a short note on FDA-approved algorithms in health care.	L-3	CO4	**05**
5.	**a.**	In what ways does AI help the power industry improve its efficiency? Explain.	L-3	CO5	**05**
		OR			
	b.	What are transfer functions? Explain the advantages of using transfer functions.	L-3	CO5	**05**
6.	**a.**	How does AI improve the packing and sorting strategy in the food sector? Explain.	L-4	CO6	**05**
		OR			
	b.	Explain the concept of food supply chain optimization using ML.	L-4	CO6	**05**

7.	a.	Explain the concept of data analysis and data analytics. How is it used in AI applications?	L-3	CO1	05
		OR			
	b.	Which performance metrics measure the efficiency of AI algorithms? Explain.	L-2	CO1	05
		PART-C	**06 X 10=60**		
8.	a.	What is the difference between Artificial Intelligence, Machine Learning, and Deep Learning? What is an expert system? What are the characteristics of an expert system?	L-2	CO1	10
		OR			
	b.	What are the different types of learning methods in AI? How do these learning methods help in developing solutions for the agriculture sector?	L-3	CO4	10
9.	a.	Explain the concept of E-Nose Systems. How does this system contribute to AI in enhancing food applications?	L-4	CO6	10
		OR			
	b.	Explain the overall AI model development in the food sector/applications. What should be the basis for choosing an AI algorithm to develop applications?	L-4	CO6	10
10.	a.	How does AI resolve the challenges in health data management? Explain the dataset preprocessing steps involved in it.	L-3	CO4	10
		OR			
	b.	Compare the traditional diagnostic process versus AI-enabled diagnostic procedure. Point out some of the disadvantages of AI-enabled diagnostic techniques.	L-3	CO4	10
11.	a.	How can AI models help in early disease identification, and what data sources are typically used for this purpose?	L-3	CO3	10
		OR			
	b.	Mention some of the popular machine learning algorithms. Explain any one that is widely used in developing agriculture applications.	L-2	CO2	10
12.	a.	Explain the role of AI in power and frequency variation control. Explain the transfer function model for a two-area system.	L-3	CO5	10
		OR			
	b.	Explain the concept of Pi and Pid (Traditional T echnique) and Fuzzy techniques used in power and frequency variation control. Which technique is more beneficial? Explain.	L-3	CO5	10
13.	a.	Describe the disease detection process in soybean pants using the AI technique. What are the advantages of this overall process?	L-3	CO2	10
		OR			
	b.	Explain the concept of a Computer Vision system (CVS). How is it used to develop AI-enabled applications?	L-3	CO5	10